

```
In [5]: import seaborn as sns
premiums = [333, 320, 301, 280, 270, 251, 260, 255, 355, 345, 355,
sns.distplot(premiums)
```

/var/folders/md/dysjhlbs7nxgp61th7x6hm5m0000gn/T/ipykernel_93869/2097640001.py:3: UserWarning:

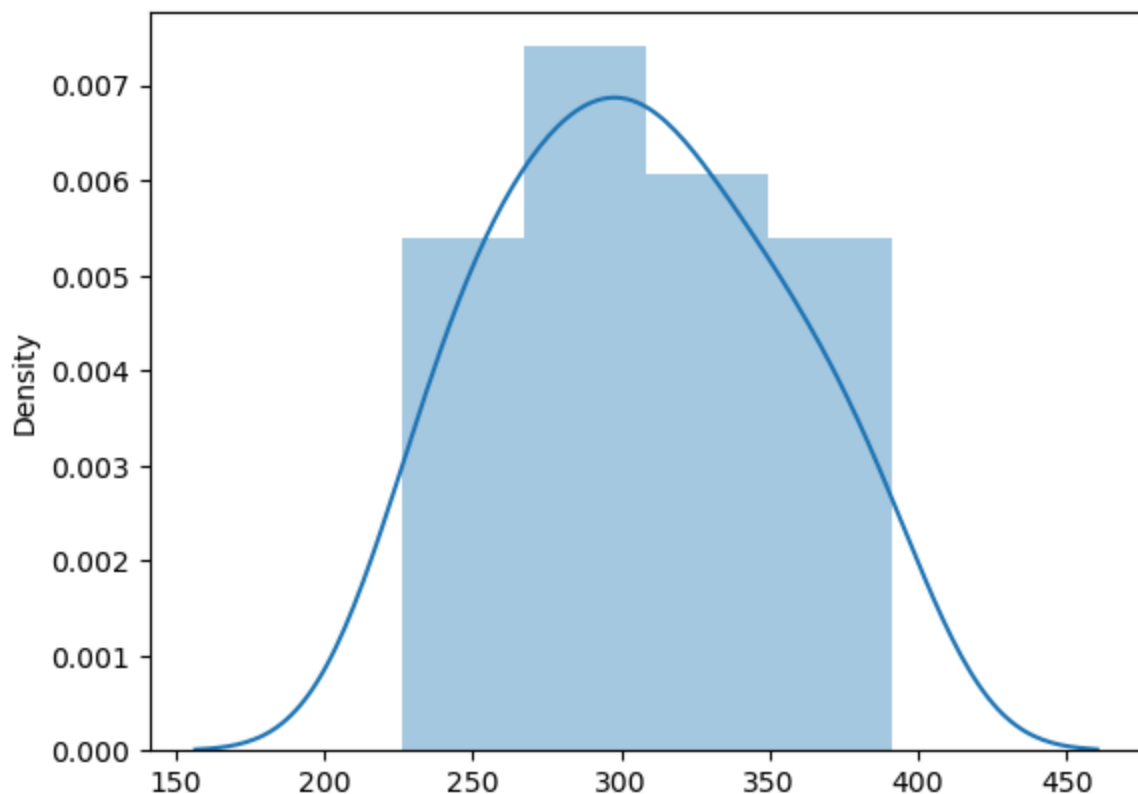
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(premiums)
```

Out[5]: <Axes: ylabel='Density'>



```
In [9]: from scipy.stats import normaltest
normaltest(premiums)
```

Out[9]: NormaltestResult(statistic=2.8425253227965745, pvalue=0.24140900654140013)

```
In [17]: #h0 serem iguais/equivalentes
#h1 serem diferentes
#teste bicaudal
from scipy.stats import norm
from numpy import sqrt
u=500
```

```

n=50
conf=0.95
sig=1-conf
z_ref = norm.ppf(conf+sig/2)
print('z ref',z_ref)
x=503.24
s=4.48
z=(x-u)/(s/sqrt(n))
print('z cal' ,z)

```

z ref 1.959963984540054
z cal 5.113897256795581

```

In [27]: p_valor = (1-norm.cdf(z))*2
         print(p_valor)

```

3.155790497810784e-07

```

In [39]: from statsmodels.stats.weightstats import ztest
         amostra = [509, 505, 495, 510, 496, 509, 497, 502, 503, 505,
                    501, 505, 510, 505, 504, 497, 506, 506, 508, 505,
                    497, 504, 500, 498, 506, 496, 508, 497, 503, 501,
                    503, 506, 499, 498, 509, 507, 503, 499, 509, 495,
                    502, 505, 504, 509, 508, 501, 505, 497, 508, 507]
         ztest(x1=amostra,value=u)

```

Out[39]: (5.109559775991873, 3.229103172445772e-07)

```

In [45]: # h0 u<=37
         # h1 u>37
         from scipy.stats import t
         from numpy import sqrt, mean, std
         n=25
         u=37
         conf=0.95
         sig=1-0.95
         t.ppf(conf, n-1)

```

Out[45]: 1.7108820799094278

```

In [47]: amostra = [37.27, 36.42, 34.84, 34.60, 37.49,
                    36.53, 35.49, 36.90, 34.52, 37.30,
                    34.99, 36.55, 36.29, 36.06, 37.42,
                    34.47, 36.70, 35.86, 36.80, 36.92,
                    37.04, 36.39, 37.32, 36.64, 35.45]
         x=mean(amostra)
         s=std(amostra)
         t_calc = (x-u)/(s/sqrt(n))
         t_calc

```

Out[47]: -3.9568374713288734

```

In [51]: 1-t.cdf(t_calc,n-1)

```

Out[51]: 0.9997062063779475

```
In [55]: from statsmodels.stats.weightstats import DescrStatsW
teste_t = DescrStatsW(amostra)
teste_t.ttest_mean(value=u, alternative='larger')
```

```
Out[55]: (-3.876893119952044, 0.9996406170303819, 24.0)
```

```
In [91]: homens = [300, 1000, 4000, 5000, 300, 350, 1200, 1200, 1100, 1800, 700, 5000]
mulheres = [199788, 1200, 6000, 1000, 788, 1100, 1900, 1577, 900, 950, 1200,
xH = mean(homens)
sH = std(homens)
xM = mean(mulheres)
sM = std(mulheres)
n=len(homens)
print('Homem:', xH, sH, n)
print('Mulher:', xM, sM, n)
```

```
Homem: 2142.608 2545.5014763963504 500
Mulher: 1755.528 9002.772477476927 500
```

```
In [83]: #h0 xH-xM <= 0
#h1 xH-xM > 0
conf=0.99
sig=1-conf
norm.ppf(conf)
```

```
Out[83]: 2.3263478740408408
```

```
In [85]: num = (xH-xM) - 0
den = sqrt((sH**2/n) + (sM**2/n))
z=num/den
z
```

```
Out[85]: 2.4687064510127734
```

```
In [87]: 1 - norm.cdf(z)
```

```
Out[87]: 0.006780120310526994
```

```
In [89]: from statsmodels.stats.weightstats import CompareMeans
from statsmodels.stats.weightstats import DescrStatsW
comparador = CompareMeans(DescrStatsW(homens), DescrStatsW(mulheres))
comparador.ztest_ind(alternative='larger', value=0)
```

```
Out[89]: (2.466236508972639, 0.006827057249262632)
```

```
In [ ]:
```